

Om Bhatt

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Education

Texas A&M University

Bachelor of Science in Mechatronics Engineering, Minor in Embedded Systems

College Station, TX | May 2028

Organizations: IEEE TAMU, ASME TAMU, TURTLE Robotics, Texas Aimbots, SOMTECH, TAMUHACK, SASE

Skills

Robotics & Autonomy: ROS 2, PX4, MAVROS, SLAM, OpenCV, PWM/PID Control, Teleoperation, Gazebo, RViz, Point Clouds

Hardware and Prototyping: ESP32, Arduino, Raspberry Pi, STM32, FPGA, LiDAR, Cameras, PCB Design, Soldering, Analog Electronics, Oscilloscope & DMM Testing, Data Acquisition Systems

Programming Languages: Python, C/C++, Embedded C, Linux, Shell (Bash/Zsh), LaTeX, HTML, CSS

Simulation and CAD: SolidWorks, FEA, Autodesk Inventor, Onshape, Bambu Studio, Eagle, Multisim, MATLAB/Simulink

AI & Software Tools: Git/Github, Visual Studio Code, Workspace, Microsoft Office, LLM Integration (ChatGPT, Claude, Gemini)

Relevant Experience

R&D Robotics Engineer | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Sep 2025 – Present

- Researched and programmed autonomous flight algorithms (offboard_control_square_2) using PX4 and MAVROS over ROS 2 Humble to command UAVs for a 10-member team's autonomous UAV swarm platform assisting first responders during structural fires.
- Co-authored and presented the "Disaster Response Observation Network (DRON)" conference poster (Apr 2026).

Hardware Team Member | *Texas Aimbots, Texas A&M University*

College Station, TX | Sep 2025 – Present

- Designed 4+ structural components, including the gimbal housing, in Onshape CAD for Sentry, a fully autonomous ground robot capable of traversing the field, engaging targets, self-healing, and operating without driver input.
- Integrated the NVIDIA Jetson Orin, LiDAR, and camera systems to enable real-time perception and autonomous decision-making ahead of the 2026 ARC Championship, an international collegiate robotics competition.

AEOP Summer Robotics Research Intern | *Center for Autonomy, UT Austin*

Austin, TX | Jun 2025 – Aug 2025

- Researched autonomous off-road navigation and drift/jump control dynamics under Dr. Ufuk Topcu and Dr. Christian Ellis to architect a low-cost, 1/16 scale autonomous RC car as a modular research platform for drift and jump navigation studies.
- Executed the complete electromechanical assembly by designing and soldering custom circuit boards to successfully power and integrate all components, including a Raspberry Pi 5/Jetson Nano and motor controllers.
- Programmed a robust low-level controller in a C++/ROS 2 software stack to achieve precise Ackerman steering and throttle control for the vehicle.
- Established full teleoperation capabilities and utilized the F1Tenth simulator to validate SLAM and control algorithms with IMU fusion for post-jump re-localization and improved drift control, preparing the platform for fully autonomous operation.

Mechatronics Research Intern | *Mechatronics Vehicle Systems Lab, University of Waterloo*

Waterloo, ON | Jun 2021 – Sep 2021

- Worked under Professor Amir Khajepour on MVS Lab's autonomous shuttle, WATonoBus.
- Designed, fabricated, and programmed an LED display module for the WATonoBus autonomous shuttle to inform non-ego agents of the shuttle's intentions.
- Completed 60+ hours of hands-on development, earning a merit-based scholarship from MVS Lab.

Projects

T.U.R.T.L.E Robotics Hatchling Robot | *ESP32, Arduino, C/C++, SolidWorks, Kinematics, Actuation & Motor Control* Jan 2025 – Sep 2025

- Won 2nd Place out of 30+ competing teams at the 2025 TURTLE Robotics Hatchling Competition.
- Designed and CADded key mechanical components in SolidWorks, including a forklift-style lift and gripper mechanism engineered for rapid block acquisition.
- Programmed an ESP32 and Arduino to control precise vehicle navigation and operate the multi-stage lift, enabling the robot to successfully retrieve and stack multiple blocks.

EcoTrack: Dual-axis Solar Tracker | *ESP32, C/C++, SolidWorks, FEA, Thermo-responsive polymers, Kinematics*

Sep 2022 – Mar 2024

- Researched phototropism and thermo-responsive polymers to engineer a solar tracker with no electrical tracking components, and built an ESP32-controlled benchmark system, outperforming traditional trackers by 87% and fixed-panel systems by 119% in net energy harnessed.
- Achieved 1st place at the Austin Regional Science Fair, advanced to TXSEF State Fair, and won a \$2,000 Jacobs Engineering Scholarship and the RICOH Sustainable Development Award.

Volunteering & Leadership

Autodesk Ambassador Program | *Autodesk, Texas A&M University*

College Station, TX | Jan 2026 – Present

- Selected for a competitive ambassador program to represent Autodesk, promoting CAD tools through campus events and workshops.

Design Review Officer | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Sep 2025 – Present

- Evaluate system requirements, CAD/electronics documentation, and BOMs across 20+ projects over 3 Design Review sessions.

TXSEF Science and Engineering Fair Judge | *Texas A&M College of Engineering*

College Station, TX | Sep 2025 – Present

- Evaluate 20+ student research projects annually in the Energy: Sustainable Materials & Designs and Robotics & Intelligent Machines categories at the Texas Science and Engineering Fair (TXSEF).

Sponsorship Committee Member | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Sep 2025 – Jan 2026

- Secured corporate sponsorships for TURTLE through a new Corporate Sponsor Outreach Initiative by developing and executing outreach strategies to engage companies and fund workshops, social events, and advanced technical projects.